Java Curriculum

1. Threading File Handling:
- Creating and managing threads
- Synchronization and locks
- Thread safety and concurrent collections
- Executors and ThreadPoolExecutor
- Java 8+ features for concurrent programming (CompletableFuture, CompletableFuture API)
- Reading and writing files
- BufferedReader, BufferedWriter, FileReader, FileWriter classes
- Serialization and deserialization (ObjectInputStream, ObjectOutputStream)
- NIO (New I/O) package and Channels

2. **Java Generics and Annotations:**
- Generic classes and methods
- Wildcards and bounded types
- Generic collections (ArrayList<T>, HashMap<K, V>, etc.)
- Built-in annotations (Override, Deprecated, SuppressWarnings, etc.)

3. **Java Memory Management:**
- Garbage Collection (GC) and memory allocation
- JVM memory structure (Heap, Stack, Metaspace)
- Memory leaks and performance optimization

4. **Java 8 Features:**

Lambda Expressions:
Functional Interfaces:
Stream API:
Default Methods in Interfaces:
Static Methods in Interfaces:
Optional Class:
Date and Time API (java.time):
Method References:
Parallel and Asynchronous Programming:
Base64 Encoding and Decoding:

5. **Java 17 Features:**
- Records (a compact way to declare data classes)
- Sealed classes and interfaces
- Pattern matching for switch statements
- Text blocks (multi-line strings)
- Foreign function and memory API (JEP 391)

6. **Java Unit Testing:**
- JUnit framework for writing and running unit tests

- Mocking frameworks (Mockito, PowerMockito)

7. **Java Build Tools:**
    - Maven or Gradle for project management and build automation
    - Dependency management and plugins

8. Version Control Systems:
    - Git basics (commits, branches, merging, rebasing)
    - GitHub or GitLab for collaborative development

9. Software Development Lifecycle (SDLC):
    - Agile methodologies (Scrum, Kanban)
    - Continuous Integration and Continuous Deployment (CI/CD) pipelines

Spring Curriculum

1. **Spring Beans and Container:**
   - Bean lifecycle management (bean scopes, initialization, destruction)
   - BeanPostProcessor and BeanFactoryPostProcessor
   - Property sources and placeholders
   - Bean definition and configuration
   - Bean wiring (autowiring, explicit wiring)
   - Spring Expression Language (SpEL)

2. **Aspect-Oriented Programming (AOP):**
   - AOP concepts (aspects, join points, pointcuts, advice, weaving)
   - Creating and using aspects in Spring
   - AOP proxying mechanisms (JDK dynamic proxies, CGLIB proxies)

3. **Spring Data Access:**
   - JDBC template and data access objects (DAOs)
   - ORM support with Spring (Hibernate, JPA)
   - Transactions management (declarative transactions, @Transactional)

4. **Spring Boot:**
   - Introduction to Spring Boot and its features
   - Auto-configuration and starters
   - Embedded servers (Tomcat, Jetty, Undertow)
   - Spring Boot Actuator for monitoring and managing applications
5. **Spring Security:**
   - Authentication and authorization concepts
   - Securing web applications with Spring Security

- User authentication (form-based, LDAP, OAuth)
- Role-based access control (RBAC)

6. **Spring RESTful Web Services (Spring Web MVC):**
    - Building RESTful APIs using Spring MVC
    - @RestController and @RequestMapping annotations
    - Content negotiation (JSON, XML)
    - Exception handling in RESTful services

7. **Spring Cloud (Optional for Microservices):**
    - Overview of microservices architecture
    - Spring Cloud components (Service Discovery, API Gateway, Config Server) - Distributed tracing and monitoring with Spring Cloud Sleuth and Zipkin

8. **Integration with Other Frameworks and Technologies:**
    - Integrating Spring with other frameworks (e.g., Apache Camel, JMS, Kafka)
    - Using Spring with frontend frameworks (Angular, React, Vue.js)

9. **Deployment and DevOps with Spring:**
    - Packaging and deploying Spring applications
    - Continuous Integration and Continuous Deployment (CI/CD) pipelines for Spring projects
    - Docker and containerization for Spring applications

10. **Monitoring and Management:**
    - Monitoring Spring applications with Spring Boot Actuator and Micrometer
    - Logging strategies and tools (Logback, Log4j2)

Hibernate Curriculum

1. **Introduction to ORM:**
    - Overview of Object-Relational Mapping (ORM)
    - Benefits of using ORM frameworks like Hibernate

2. **Entity Mapping:**
    - Mapping Java classes (entities) to database tables
    - Primary key generation strategies (e.g., @GeneratedValue)
    - Mapping properties (fields and associations) using annotations or XML
3. **CRUD Operations:**
    - Performing Create, Read, Update, and Delete operations using Hibernate

- Entity lifecycle (transient, persistent, detached, removed)
- Hibernate APIs for data manipulation (save, update, delete, get, load)

4. **Querying with Hibernate:**
   - HQL (Hibernate Query Language) basics
   - Criteria API for dynamic queries
   - Native SQL queries and named queries

5. **Hibernate Relationships:**
   - One-to-One, One-to-Many, Many-to-One, and Many-to-Many relationships
   - Fetch strategies (eager loading, lazy loading)
   - Cascade operations and orphan removal

6. **Caching Strategies:**
   - Hibernate caching mechanisms (first-level cache, second-level cache)
   - Configuring cache providers (Ehcache, Infinispan, Hazelcast) - Cache eviction policies and tuning options